

GOSIM Paris 2026

May 5-6, 2026 Station F, Paris



Building Navigation Foundation Models with Scalable Simulation Data

Wenzhe Cai

Shanghai AI Laboratory



CONTENTS

GOSIM Paris 2026

Part 01

Navigation Data Bottleneck

Part 02

Navigation System-1: NavDP

Part 03

Navigation System-2: InternVLA-N1

Part 04

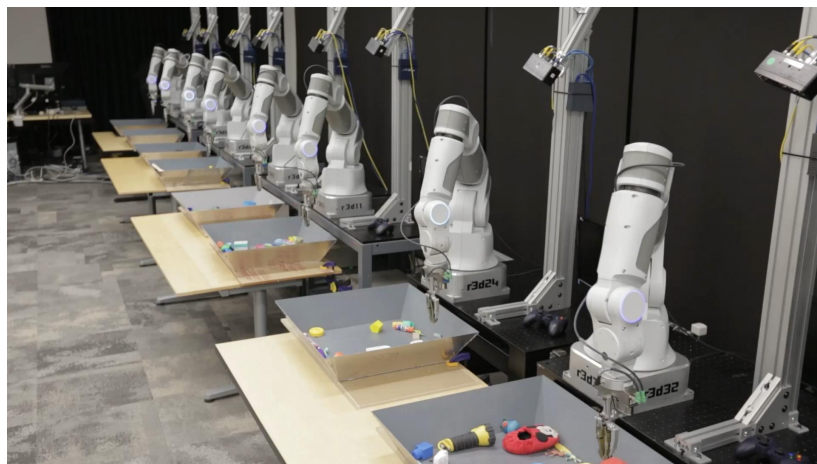
Conclusion & Takeaways



01

Navigation Data Bottleneck





Manipulation:

Easier Task Reset

Centralized Data Factory

Small Embodiment Gap

Oracle Policy



Navigation:

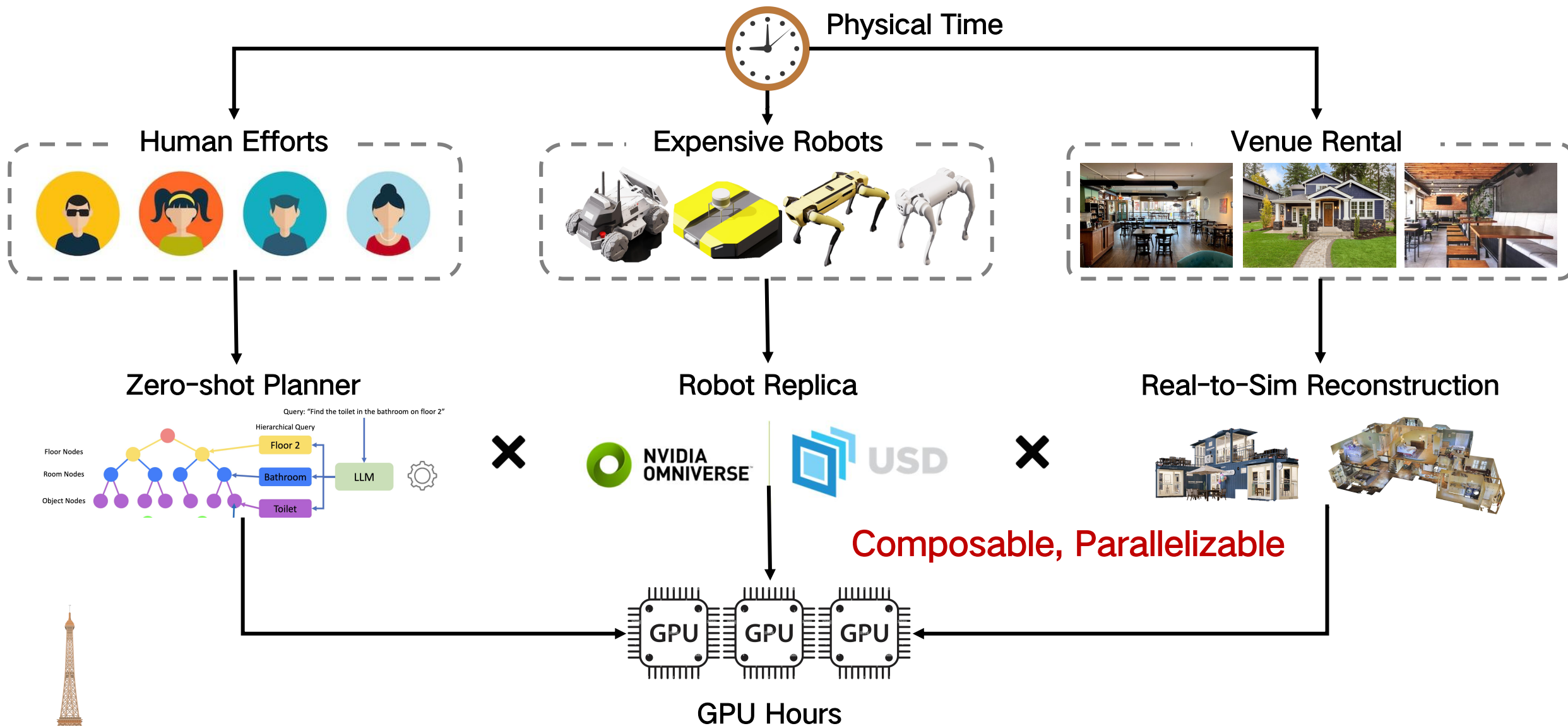
Difficult Scene Reset

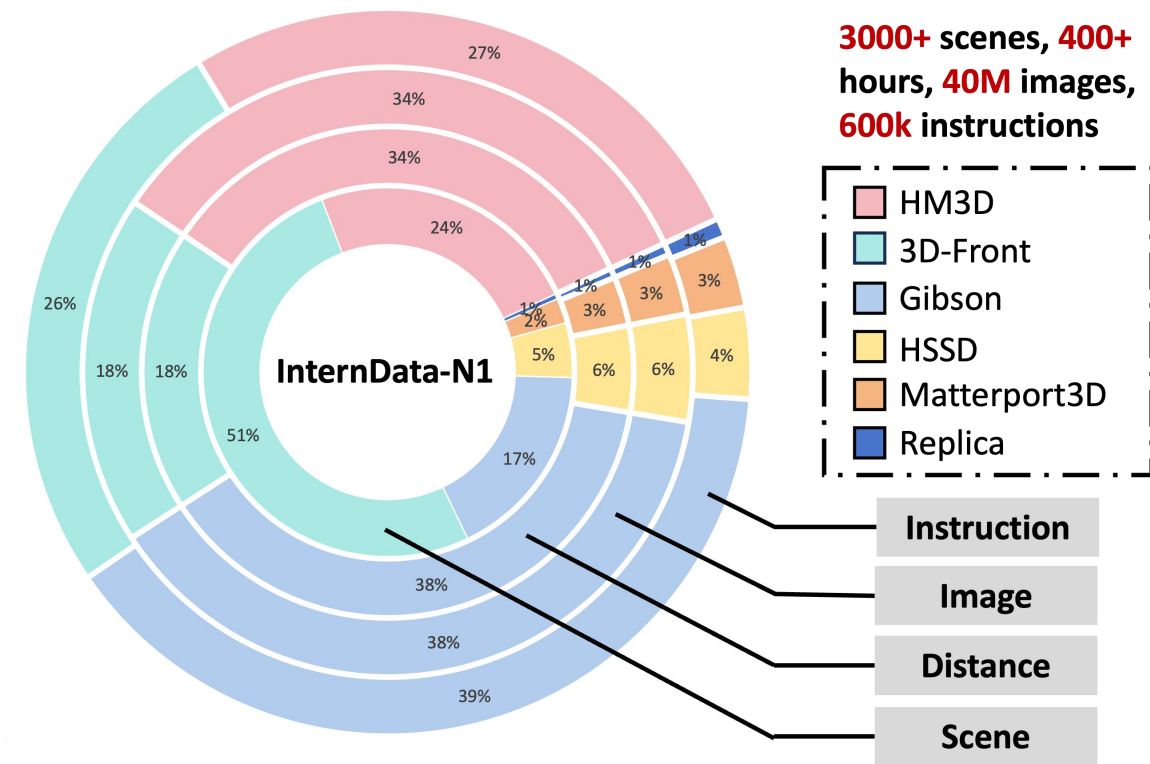
Standalone Teleoperation

Variant Mobile Base

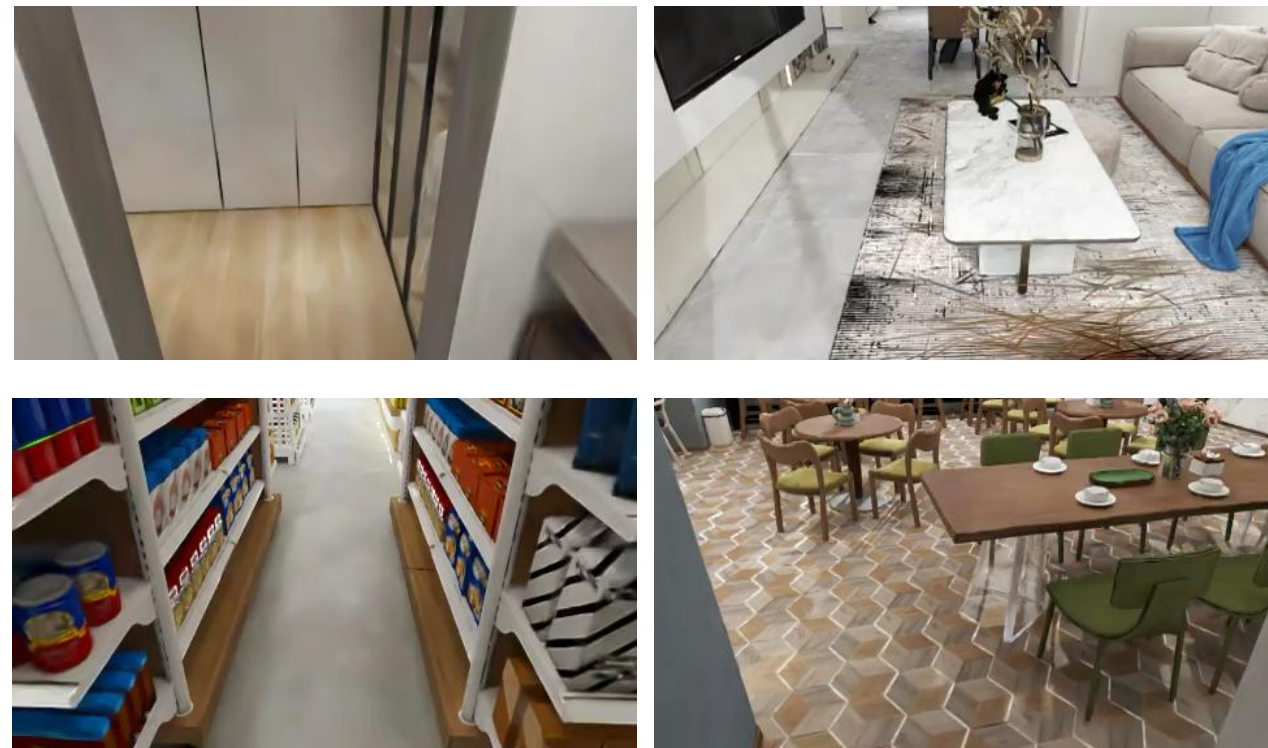
Oracle Policy



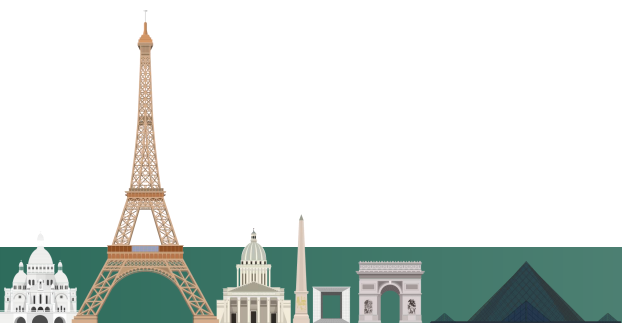


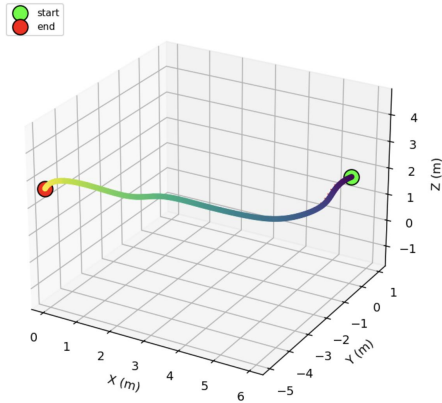


Dataset Summary



Dataset Example





Camera Poses



RGB-D Video

Instruction: Start from the long dining table in front of you, walk forward through the dining area with round tables equipped with wooden chairs with green cushions...

Annotated Instructions

Navigation Foundation Model



02

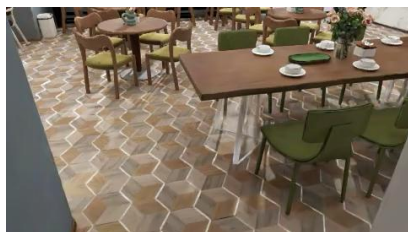
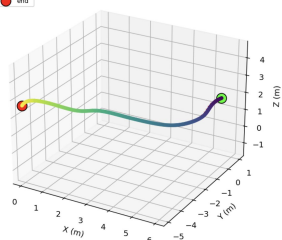
Navigation System-1: NavDP



Simulation / Real Data

$\langle \text{observation}, \text{action} \rangle$

pairs



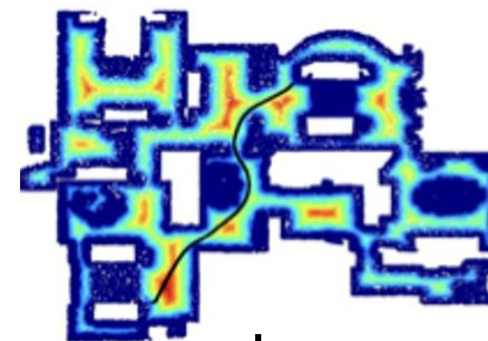
Diffusion Policy
(Action Head)

Goal-Oriented
Trajectory Generation

**Safe
Navigation
Behavior**

Simulation Privileged Information

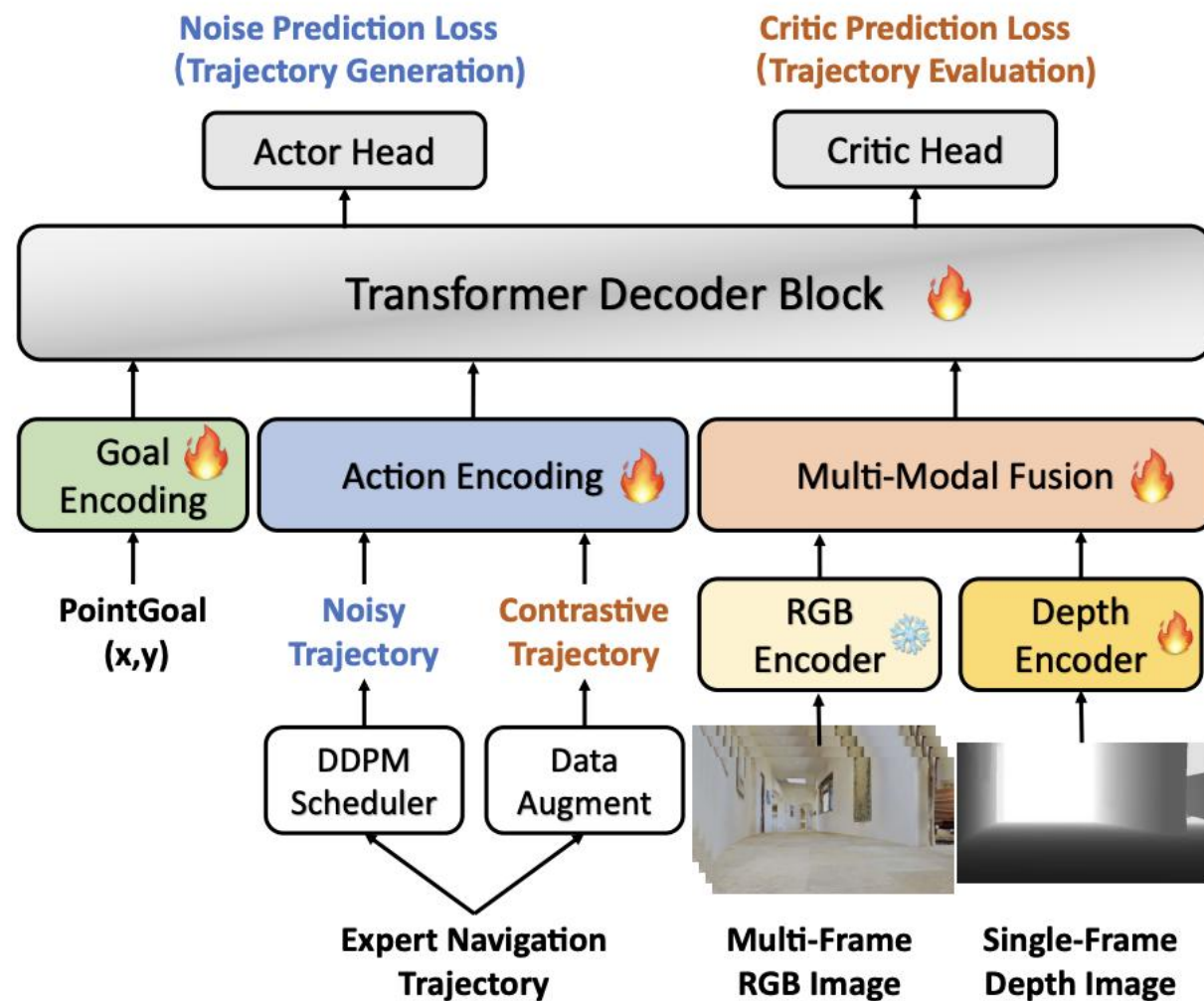
Additional Supervision



Score Function
(Critic Head)

Goal-Agnostic
Trajectory Selection





Summary:

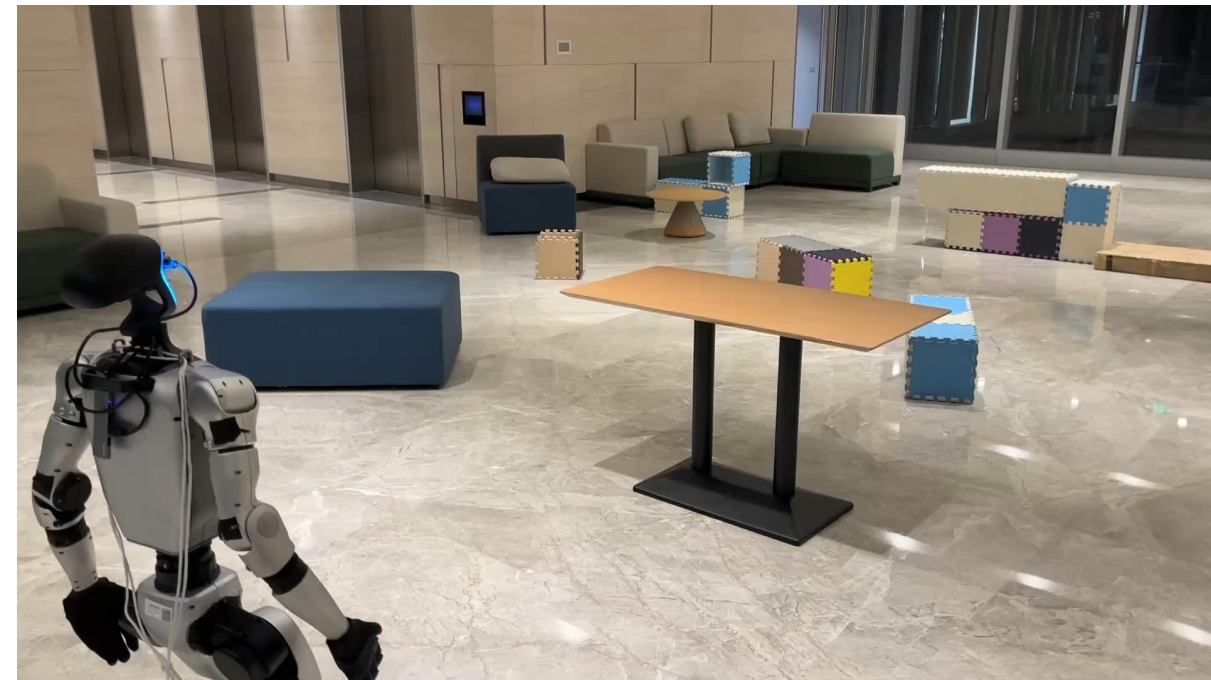
- Depth Input $\rightarrow$ Metric-scale Prior
- Multi-Frame Input $\rightarrow$ Dynamic Obstacles
- Frozen DepthAnything V2 RGB Encoder



Robustness to Dynamic Obstacles



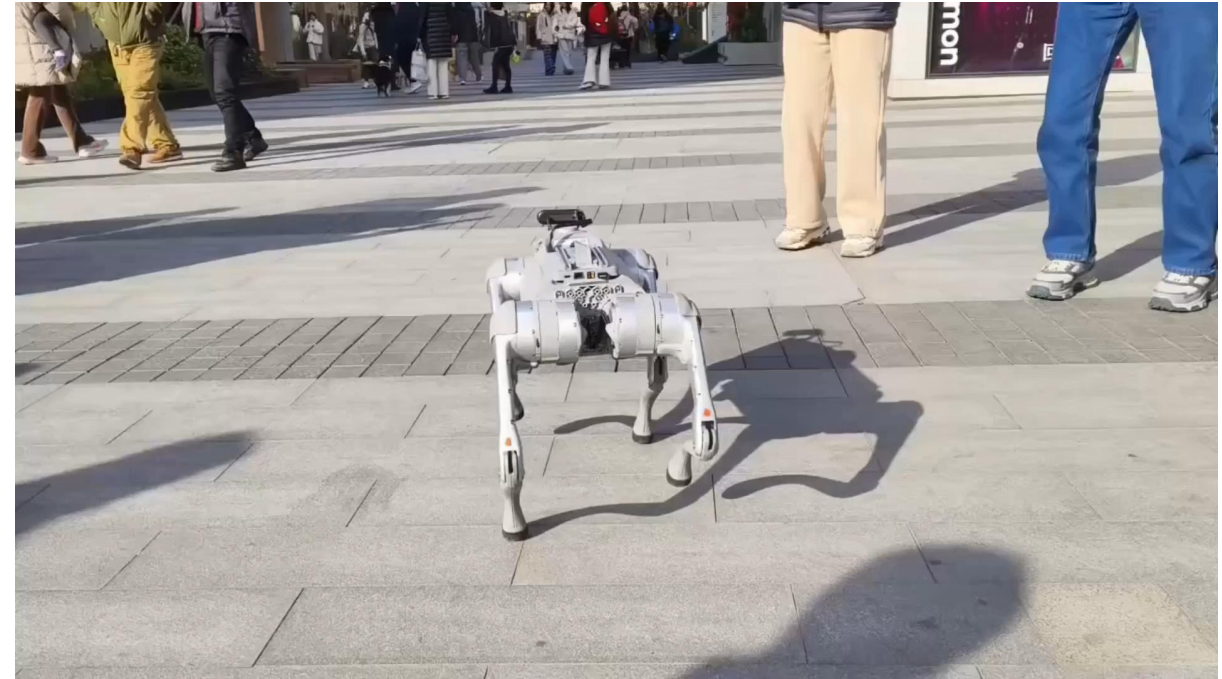
Cross-Embodiment Generalization



Locomotion-Policy Adaptation



Outdoor Generalization



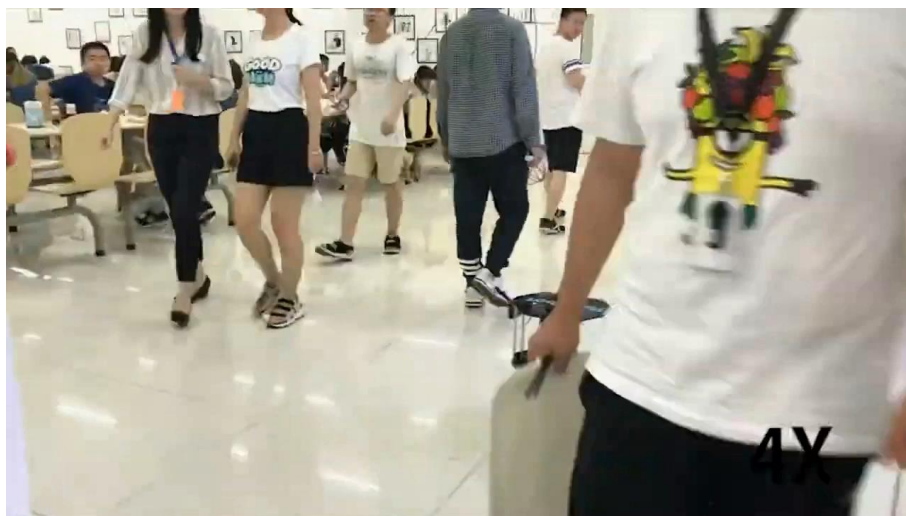
02

Navigation System-2: InternVLA-N1

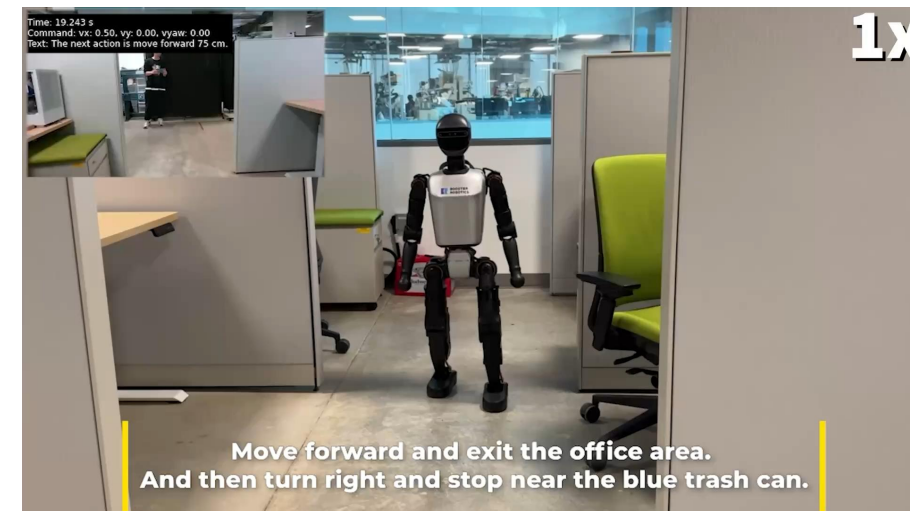


	Qwen2.5-72B Base	Qwen2.5-Plus Base	Llama-4-Maverick Base	DeepSeek-V3 Base	Qwen3-235B-A22B Base
Architecture	Dense	MoE	MoE	MoE	MoE
# Total Params	72B	271B	402B	671B	235B
# Activated Params	72B	37B	17B	37B	22B
<i>Math & STEM Tasks</i>					
GPQA	<u>45.88</u>	41.92	43.94	41.92	47.47
GSM8K	91.50	<u>91.89</u>	87.72	87.57	94.39
MATH	62.12	62.78	<u>63.32</u>	62.62	71.84
<i>Coding Tasks</i>					
EvalPlus	65.93	61.43	<u>68.38</u>	63.75	77.60
MultiPL-E	58.70	62.16	57.28	<u>62.26</u>	65.94
MBPP	<u>76.00</u>	74.60	75.40	74.20	81.40
CRUX-O	66.20	68.50	<u>77.00</u>	76.60	79.00

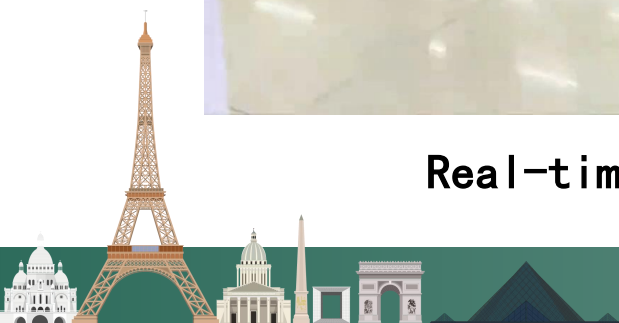
Larger Models often top the Reasoning benchmarks



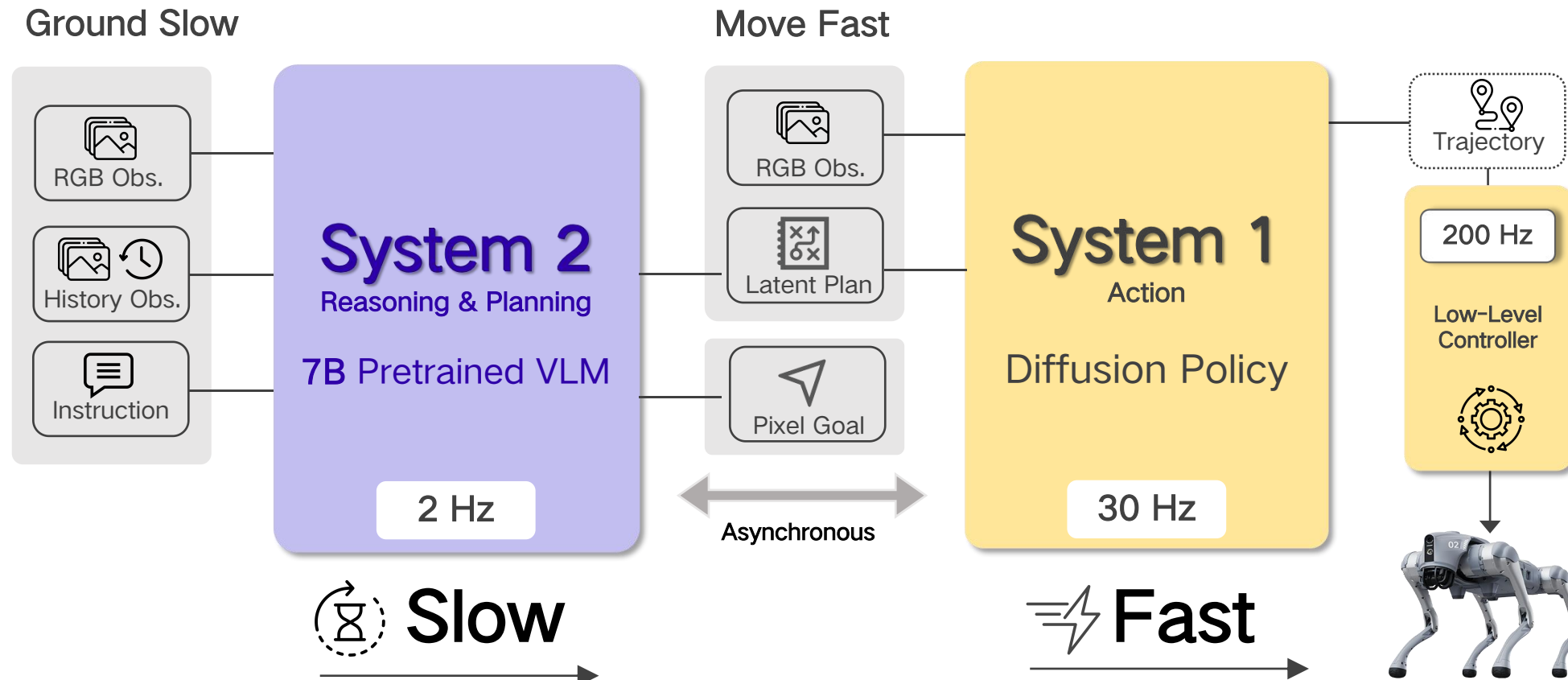
Real-time Inference

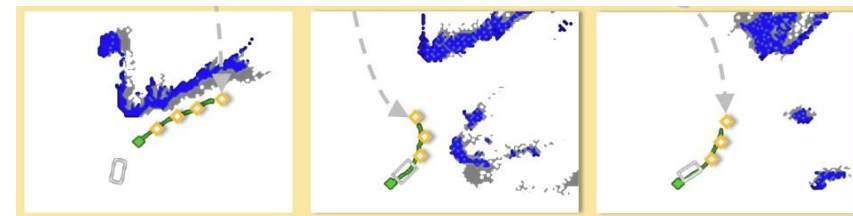
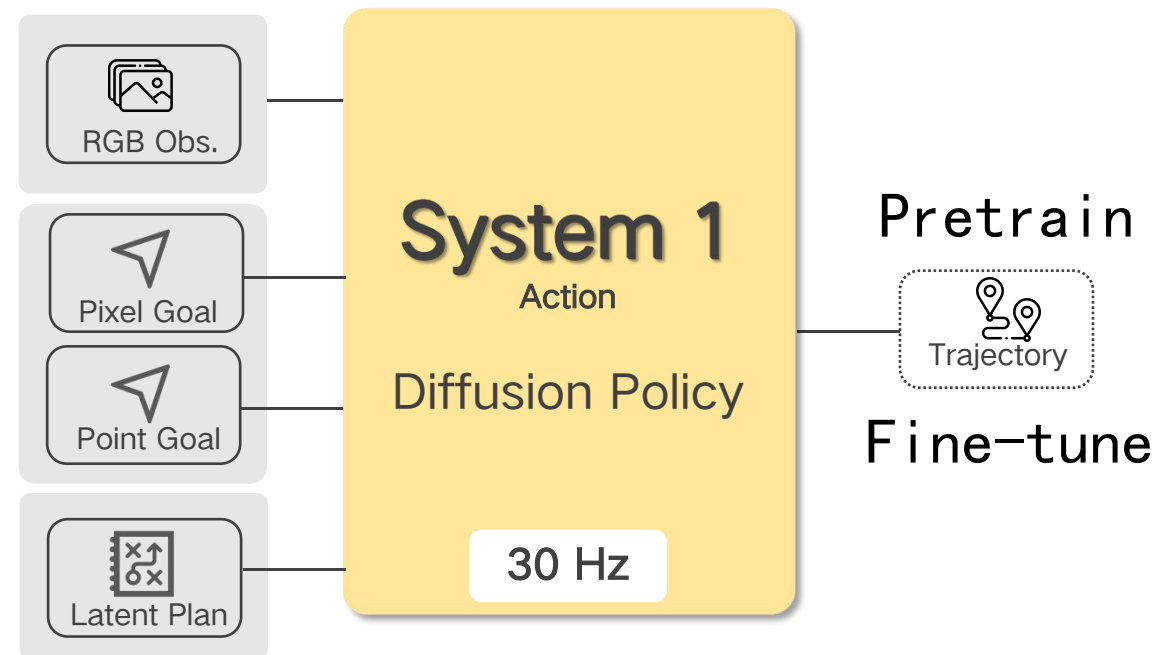
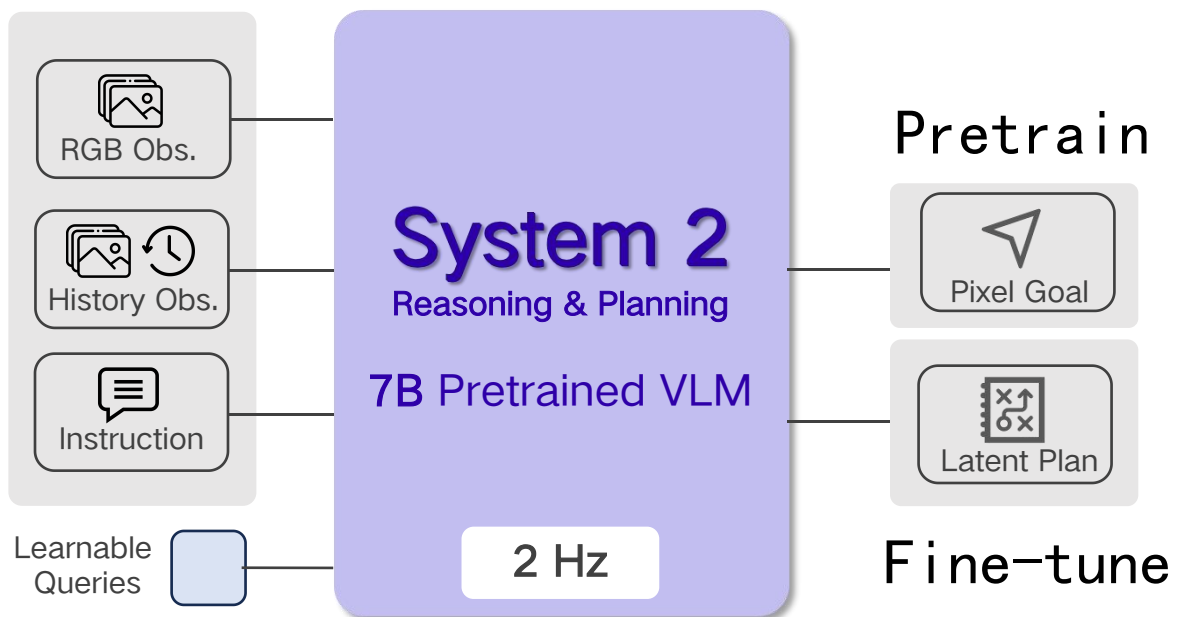


LLM-Based Next-Action Token Predictions cannot satisfy both requirements



- **System2**: Multi-Modal Perception & Reasoning
- **System1**: Real-time Obstacle Avoidance and Path-Planning







Instruction: Walk straight ahead, turn right after seeing the billiard table, and enter the room with shelves.



Instruction: Walk through the bamboo forest and turn left. Reach a man sitting on the sofa





***Instruction:** Walk out of this house and find a trash bin.*



***Instruction:** Walk towards the orange coffee sculpture and go upstairs. Then keep walking straight and turn right at the end. After that turn left and walking to the man with a black umbrella. Stop in front of the doors with red handles.*



Problems that can be solved by our Sim2Real pipeline:

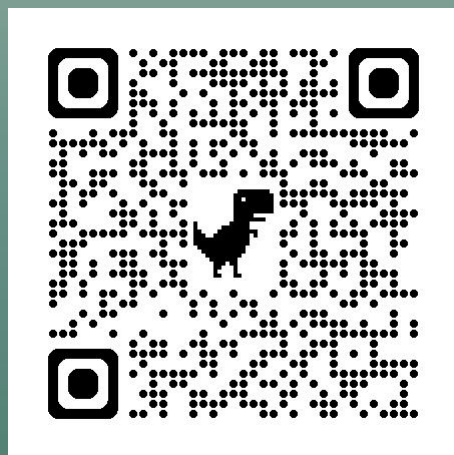
- 2D trajectory planning (obstacle avoidance)
- Step-by-step Navigation Instruction Following
- How to build scalable data flywheel is the central problem (Video-to-Scene)

Problems Requires Future Works:

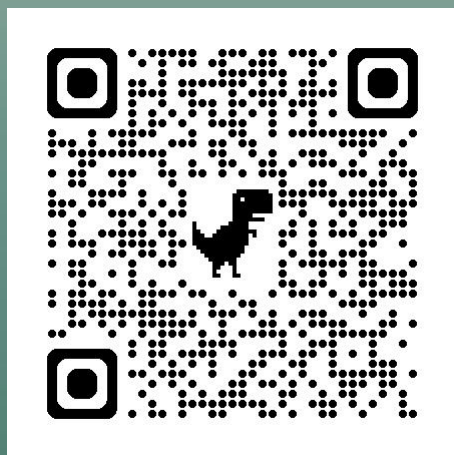
- 3D Navigation Planning (Affordance varies across different embodiment)
- Long-Context Memory Aware Navigation Problem



Thanks!



NavDP



InternNav

